Training Day 19

1. Topics Covered and Brief Explanation

1. Introduction to OpenCV: OpenCV (Open Source Computer Vision Library) is an open-source library used for image processing, video analysis, and computer vision applications.
2. Reading Images: OpenCV provides the cv2.imread() function to load images from a file.
3. Displaying Images: The cv2.imshow() function is used to display an image in a separate window.
4. Saving Images: The cv2.imwrite() function saves an image to a specified file path.
5. Understanding Image Properties: An image can be represented as a NumPy array. Its shape provides information about height, width, and colour channels.
6. Image Reading Modes: OpenCV supports different image-reading modes:
   - cv2.IMREAD_COLOR – loads a colour image.
   - cv2.IMREAD_GRAYSCALE – loads an image in grayscale.
   - cv2.IMREAD_UNCHANGED – loads an image with its original channels, including an alpha channel if present.

2. Tools and Technologies Used

Tool/Technology	Purpose
Python	Programming language
OpenCV	Image reading and processing
NumPy	Image data representation
VS Code / Google Colab	Code execution

3. Practical Learning / Code Examples

A basic program was used to read an image, display its properties, and save a copy.

```
import cv2

# Read an image

image = cv2.imread("image.jpg")

if image is not None:

    print("Image Shape:", image.shape)

    print("Image Size:", image.size)

    # Display image

    cv2.imshow("Original Image", image)
```

```
    # Save image
    cv2.imwrite("saved_image.jpg", image)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
else:
    print("Image not found")
```

Explanation:

- cv2.imread() reads the image.
- image.shape returns its dimensions.
- image.size returns the total number of pixel values.
- cv2.imshow() displays the image.
- cv2.imwrite() saves the image.
- cv2.waitKey(0) waits for a key press before closing the window.

4. Learning Outcomes

- Understood the purpose and features of OpenCV.
- Learned how to read and display images.
- Practised saving images using OpenCV.
- Understood image dimensions and pixel representation.
- Explored different image-reading modes.